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Characteristics and influencing factors of the industry-university-research collaborative innovation network in China's new energy vehicle industry

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ABSTRACT

Industry-university-research collaboration is the primary channel of industrial innovation, and China is a common context for research on innovation in the new energy vehicle industry. This study analyzes the industry-university-research collaborative innovation network in China's new energy vehicle industry using various software programs, including Gephi and ArcGIS, based on collaborative invention patent data related to the new energy vehicle industry provided by the global patent database incoPat. According to the findings of this study, the highest number of patents obtained through industry-university-research collaboration was observed in new energy vehicle device and accessory manufacturing, whereas the number of new entrants was significantly higher than the number of incumbents albeit in a downward trend, suggesting that the creation of new collaborative relationships among new entrants has always been the main phenomenon in the process of building an industry-university-research innovation system in the new energy vehicle industry. As private firms have long been playing an integral role in industry-university-research collaboration in the new energy vehicle industry, the number of private firms in the network exhibited an upward trend annually, while the dual heterogeneous connection mode was the primary connection mode in industry-university-research collaboration. There were some differences between the spatial characteristics of intraprovincial and interprovincial collaboration in the industry-university-research collaborative network in China's new energy vehicle industry. The analysis results obtained using GeoDetector revealed that the influence of economic base, level of openness to the outside world, government dominance, market dominance, regional science and technology finance, and entrepreneurial spirit varied from one stage to another.

1. Introduction

As China's energy reserves are characterized by more coal but less oil and gas, China encounters a series of energy security issues arising from its huge demand for oil and heavy reliance on imports. To reflect the country's determination and resolve toward green development, China has pledged to achieve carbon peak and carbon neutrality by 2030 and 2060, respectively. With the development of the new energy vehicle (NEV) industry, China will not only be able to lower carbon emissions and reduce its reliance on oil to a greater extent [1–3] but also utilize this industry to overtake other countries in the automotive industry.

The development of any emerging industry is inseparable from innovation. Hence, innovation also plays a pivotal role in developing this knowledge and technology-intensive industry [4–6]. While firms, universities, and research institutions, the key entities of innovation activities, play a key role in different capacities in technological innovation, industry-university-research collaboration (IURC) arising from the collaboration between these entities is even more critical to the technological progress of the industry [7–10].

Among the studies on IURC, the first area of research concerns the theories related to IURC. In 1997, EtzKowitz and Leydesdorff proposed the triple helix model of university-industry-government relations [11], laying a solid theoretical foundation for studying IURC. From the perspective of transaction cost theory, industry-university-research

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collaborative innovation (IURCI) is essentially a transaction activity formed via the flow of knowledge between patent applicants in different fields, that is, firms, universities, and research institutions, to achieve goals in specific areas by realizing their own strengths while reducing transaction costs [12,13]. From the standpoint of social network theory, IURC is essentially a network formed by interconnections and interactions between nodes, where the interconnections between enterprises, universities, and research institutions in the network are the primary issues of concern in social network theory [14–16].

The second area of research centers upon the structural characteristics of IURC. Owing to the presence of network connection characteristics in IURC, scholars adopted the social network analysis method and drew on complex network theory to carry out related research from the viewpoint of network characteristics such as network centrality and network centralization, where they identified the presence of growth characteristics in IURC networks [17]. While further observations of the clustering coefficients and path lengths of IURC networks demonstrated the small-world and scale-free characteristics in such networks, in-depth studies also revealed that these network characteristics are conducive to knowledge diffusion in IURC networks, where the higher the efficiency of knowledge diffusion, the more efficient the innovation network [18–21]. In addition, the entry of academic and research institutions with strong innovative capabilities is the key to bolstering the benign development of IURC networks [22].

The third area of research revolves around the factors influencing IURC. After conducting research from theoretical and practical aspects, scholars have argued that IURC increases innovation levels and reduces collaboration costs. Firm size, innovation input, government financial support, level of openness, mutual trust between collaborating entities, and proximity between collaborating entities are the primary factors influencing IURCI [23–27].

In summary, as IURCI brings great benefits to the development of the NEV industry, extant studies have shifted their focus from public institutions such as universities in the early days to discussions on the importance and necessity of IURC, as well as empirical evidence on the factors influencing it [28–32]. With the deepening of research and the growing maturity of scientific and technological methods, the paradigm of network research has gradually become mainstream, while the levels of research on IURC networks have become more diverse because of social network analyses [33–35].

However, while there has been a dearth of papers analyzing IURCI in the NEV industry using joint patent application data, there have been few studies analyzing the evolution of IURC in this industry and the influencing factors over a longer period. Meanwhile, most studies on the influencing factors carried out analyses from the micro perspectives of firms, universities, and research institutions but lacked discussions from a macro perspective at the regional level. With China being a typical nation in the study of innovation in the NEV industry [36], at what stage is the IURCN in China's NEV industry? What are the characteristics of this network? Moreover, what are the factors influencing this network? Hence, this study analyzes the IURCI network in China's NEV industry from 2004 to 2019 based on the global patent database incoPat data. First, this study analyzes the fundamental characteristics of IURC in China's NEV industry by industry category, organizational characteristics, and connection mode. Next, using the social network analysis method, this study analyzes the fundamental characteristics of the provincial industry-university-research collaborative network in provinces where patent applicants are located. Lastly, this study investigates the factors influencing the provincial industry-university-research collaborative network in China's NEV industry using GeoDetector.

This study has many theoretical contributions. First of all, this study attempts to investigate and analyze the structure of the IURCI network in China's NEV industry from the standpoint of patent applicant heterogeneity and unveil the characteristics of the patterns of the network from a heterogeneity perspective, to not only offering theoretical support for the study of the evolution of the IURCI network but also providing local governments with some references when formulating related policies [37–43]. Secondly, This study analyzes in depth the position and role of each nodal province in the provincial IURCN, which not only helps to comprehend the functions and role of provinces in innovation development but also facilitates efforts to deepen theoretical research on provincial and municipal geography [44–47]. Finally, the study of the IURCI network in the NEV industry from a network perspective based on the social network analysis method is a supplement and improvement to the traditional attribute data-based research. The network perspective can reflect the interconnections between the industry, universities, and research institutions more profoundly, further deepening and expanding the theory of IURC [17–20,33–35].

2. Research data and methods

2.1. Research data

The data used in this study are sourced from the global patent database incoPat. Searches were conducted according to international patent classification for the NEV industry stipulated in the "Relationship Table of Strategic Emerging Industry Classification and International Patent Classification (2021) (Trial)" [48] published by the China National Intellectual Property Administration. First, this study excluded patents whose applicants were natural persons and foreign companies, involving only one applicant. Second, by referring firms, groups, and factories as "industry," universities and colleges as "university," and research institutes and centers as "research," this study selected patents involving IURC, where industry-university, industry-research, and university-research collaboration between patent applicants were all regarded as IURC. Third, this study excluded patents from 2020 to 2022 due to the greater impact of the COVID-19 pandemic on patent development and application over this period, where these patents not only failed to reflect the pattern of evolution of the network objectively and meet the pattern of innovation development but also did not undergo the long review cycle from application to publication. Finally, 4258 patents that met the relevant requirements between 2004 and 2019 were successfully obtained. In the measurement of patent collaboration, patent applicants were first classified into first and non-first patent applicants and then patent collaboration was divided into collaboration led by first patent applicants and collaboration participated by non-first patent applicants to form a directed data matrix of collaborative innovation between patent applicants. For instance, if four patent applicants, namely A, B, C, and D, jointly apply for a patent, A is regarded to have collaborated with B, C, and D (i.e., A-B, A-C, and A-D) once each. To present the characteristics of the evolution of the IURCN in the NEV industry as objectively as possible and avoid differences in certain unique years, this study adopted the time interval method, where the study period was divided into four four-year intervals: 2004 to 2007, 2008 to 2011, 2012 to 2015, and 2016 to 2019.

2.2. Research methods

2.2.1. Social network analysis

The method of social network analysis is based on the interaction of individuals, and the research and analysis of the relationship between them. Based on the social network analysis method, this study analyzes the changes in various indicators of the IURCN in the NEV industry, including network centrality, network density, and average path length, to unveil the characteristics of the industry-university-research innovation connection network in the NEV industry [49–53]. Network centrality is used to measure the importance or influence of nodes in the network. Density is used to measure the closeness of the relationships between nodes in a network. The higher the network density, the more point pairs that have relations between nodes in the network, that is, the closer the overall relationship between nodes in the network. The average path length is mainly used to measure the average of the shortest distance between any two nodes in the network, and can be used to reflect the efficiency of information transmission in the network. The smaller the average path length, the more efficiently information is transferred from one node to another [54].

2.2.2. GeoDetector

GeoDetector is a spatial analysis model for detecting the relationship between a geographical attribute and its explanatory factors. It is commonly used in explaining the factors influencing various types of phenomena. The advantage of this method lies in the fact that it involves only a handful of preconditions when analyzing multiple types of data [55–57]. This method is expressed by the following equation:

$$q = 1 - \frac{\sum_{h=1}^{L} \sigma_h^2 N_h}{N\sigma^2} \tag{1}$$

where q is the ability to detect the effect of the influencing factor on the centrality of the provincial industry-university-research collaborative network in China's NEV industry; h is the classification of each factor of the variable, where $h = 1, \ldots, L$; L is the grading of each factor of the variable; σ^2 is the total variance of the centrality of the provincial IURCN in China's NEV industry; σ_h^2 is the variance of the provincial IURCN in China's NEV industry; N is the number of provinces in China; and N_h is the number of types of influencing factor X. Specifically, the value of q lies in the interval [0, 1], where the higher the value of q, the greater the effect of the factor on the provincial IURCI network in China's NEV industry.

3. Characteristics of IURC in China's NEV industry

3.1. Characteristics in different categories of the NEV industry

According to the "Relationship Table of Strategic Emerging Industry Classification and International Patent Classification (2021) (Trial)" [48] issued and published by the China National Intellectual Property Administration, the NEV industry is classified into four categories (Table 1). Based on the findings of this study, the highest number of patents obtained via IURC is observed in NEV devices and accessory manufacturing, which has always demonstrated strong growth momentum. Specifically, 1356 patents from 2016 to 2019 were recorded in this industry category, three times that from 2012 to 2015. Next, the second highest number of patents obtained via IURC is observed in NEV-related facility manufacturing. Specifically, this industry category records 1084 patents from 2016 to 2019, double that from 2012 to 2015. However, the lowest number of patents obtained via IURC is observed in NEV-related services.

Table 2
Characteristics of new entrants and incumbents in the IURCN in the NEV industry.

Stage		2004 to 2007	2008 to 2011	2012 to 2015	2016 to 2019
Patent applicant	Incumbents	27	15	111	472
	New entrants	–	140	616	1394
	Proportion of new entrants (%)	–	90.32 %	84.73 %	74.71 %
First patent applicant	Incumbents	13	6	54	233
	New entrants	–	73	316	739
	Proportion of new entrants (%)	–	92.41 %	85.41 %	76.03 %
Non-first patent applicant	Incumbents	14	9	67	271
	New entrants	–	78	370	877
	Proportion of new entrants (%)	–	89.66 %	84.67 %	76.39 %

3.2. Organizational characteristics of IURC in the NEV industry

In this study, patent applicants are classified by organizational attribute into two categories, that is, new entrants and incumbents. The number and growth of both categories of patent applicants in the IURCN in China's NEV industry over the study period reflect the changes in the size of the network. As seen in Table 2, the size of the network continues to grow as there are significantly fewer incumbents than new entrants, while the proportion of new entrants declines from 90.32 % to 74.71 %. From the perspective of first and non-first patent applicants, it is evident that, be it incumbents or new entrants, the number of non-first patent applicants is higher than the number of first patent applicants; in addition, there is an overlap between both first and non-first patent applicants. These findings arise primarily because first patent applicants possess higher research and development (R&D) capabilities while assuming the primary role of developing patents. Therefore, the number of first patent applicants is lower than the number of non-first patent applicants.

As seen in Table 3, by building an industry-university-research innovation system in the NEV industry, the creation of new collaborative relationships by new entrants has always been the main phenomenon in this process, while incumbents maintain their existing collaboration and play an important role. However, there is less collaboration between incumbents and new entrants. This demonstrates some extent of path dependence in IURC in the NEV industry and the existence of factional barriers in existing collaboration. While striving to join the existing research community, new entrants also continue to expand the scope of their research teams via new collaborative relationships.

As far as organization type in Table 4 is concerned, private firms have long been playing a key role in IURC in the NEV industry, as the number

Table 1
Number of patents obtained via IURC in different categories of the NEV industry.

Category of strategic emerging industry	NEV manufacturing	NEV device and accessory manufacturing	NEV-related facility manufacturing	NEV-related services	Total
2004 to 2007	2	15	13		30
2008 to 2011	11	103	60	9	183
2012 to 2015	92	469	464	77	1102
2016 to 2019	279	1356	1084	228	2943
Total	384	1943	1621	314	4258

Table 3
Process of building an innovation system in the IURCN in the NEV industry.

Stage	A-B (Existing collaboration maintained by incumbents)	A-D (New collaboration established by incumbents and new entrants)	A-G (Collaboration with incumbents pursued by new entrants)	E-F (New collaboration established by new entrants)	Total
2004–2007	–	–	–	–	–
2008–2011	64	10	10	105	189
2012–2015	245	180	204	546	1175
2016–2019	875	699	530	1127	3231
Total	1184	889	744	1778	4595

Table 4
Types of organization in the IURCN in the NEV industry.

Stage	Private firms	Research institutions	Universities	State-owned firms	Hybrid firms	Foreign-invested firms
2004–2007	8	5	9	4	1	0
2008–2011	62	33	38	14	3	5
2012–2015	346	186	120	44	16	15
2016–2019	985	464	250	89	42	36

of private firms in this network exhibits an upward trend from year to year. Specifically, private firms appear 1401 times over the 16 years from 2004 to 2019, the highest among all types of organizations. Meanwhile, research institutions surpassed universities between 2012 and 2015 and experienced growth at a faster pace in this respect. Overall, research institutions appear 688 times over the study period, thus becoming the backbone of IURC in the NEV industry. However, universities and state-owned firms demonstrate a growing trend in this regard, as evidenced by the increasing number of universities and state-owned firms in this network annually. Universities and state-owned firms appear 417 and 151 times over the study period, ranking third and fourth among all types of organizations, respectively, suggesting that both universities and state-owned firms are vital participants in the IURCN. Lastly, foreign-invested and hybrid firms appear the least number of times over the study period, taking the last two spots in terms of IURC.

3.3. Characteristics of the evolution of connection modes in IURC in the NEV industry

As universities and research institutions are upstream organizations that engage in "knowledge creation" while firms are downstream organizations that engage in "knowledge application" with heterogeneity in knowledge attributes across the knowledge chain, knowledge distance exists between universities and research institutions and firms. However, there is institutional distance between organizations managed by the government within the system (e.g., state-owned firms and universities) and organizations not under government control outside the system (e.g., private firms) due to distinct institutional attributes between the two [37–43]. In this study, the connection modes for IURC are divided based on the measurement of knowledge distance and institutional distance into three types, namely proximity, cross-knowledge, and dual heterogeneous connection modes, as shown in Table 5.

Table 5
Characteristics of connection modes in the IURCN in the NEV industry.

Connection mode	2004–2007	2008–2011	2012–2015	2016–2019
Cross-knowledge connection mode	10	63	264	313
Proximity connection mode	9	9	150	465
Dual heterogeneous connection mode	15	117	761	2453

Based on the above analysis, it can be seen that the double heterogeneous connection mode is the primary connection mode in IURC.

(1) Proximity connection mode

This mode does not manifest two collaborating parties spanning the knowledge and institutional distance between them and involves two types of collaboration: (a) private firm-foreign-invested firm collaboration and (b) university-research institution collaboration. In the second type of collaboration, Jiangsu Huadong Institute of Li-ion Battery tops the list both in the number of invention patent applications and the number of patent collaborations as it establishes 34 connections (22.67 %) in the third stage and 54 connections (11.61 %) in the fourth stage.

(2) Cross-knowledge connection mode

This mode entails two collaborating parties only spanning the knowledge distance between them and involves two types of collaboration: (a) state-owned firm-university collaboration and (b) state-owned firm-research institution collaboration. Since the first stage, there has been a 25-fold increase in the number of state-owned firm-university collaborations and an even higher (i.e., 60-fold) increase in the number of state-owned firm-research institution collaborations over the study period. Specifically, China's main state-owned firm, State Grid Corporation of China, has established 57 connections via state-owned firm-university collaboration and 25 connections via state-owned firm-research institution collaboration over 16 years.

(3) Dual heterogeneous connection mode

This mode embodies two collaborating parties spanning both the knowledge and institutional distance between them and involves six types of collaboration: (a) private firm-university collaboration, (b) private firm-research institution, (c) state-owned firm-university collaboration, (d) foreign-invested firm-research institution collaboration, (e) hybrid firm-university collaboration, and (f) hybrid firm-research institution collaboration. Specifically, private firm-university collaboration was conspicuous over the study period as the number of such collaborations is not only the highest but also exhibits a noticeable growing trend. Private firm-university collaboration is the hot spot in the dual heterogeneous connection mode, as evidenced by 1194 connections established via this type of collaboration in the fourth stage, which is 3.22 times that in the third stage.

4. Provincial IURCN in China's NEV industry

4.1. Evolution of intraprovincial collaboration

As illustrated in Table 6, only Beijing, Guangdong, Henan, Hunan,

Table 6
Number and proportion of intraprovincial collaborations in the provincial IURCN in the NEV industry.

Province or municipality	2004–2007			2008–2011			2012–2015			2016–2019		
	Number of intraprovincial collaborations	Number of collaborations	Proportion of intrapronvincial collaborations	Number of intraprovincial collaborations	Number of collaborations	Proportion of intrapronvincial collaborations	Number of intraprovincial collaborations	Number of collaborations	Proportion of intrapronvincial collaborations	Number of intraprovincial collaborations	Number of collaborations	Proportion of intrapronvincial collaborations
Anhui							8	14	57.14 %	42	64	65.63 %
Beijing	1	9	11.11 %	7	73	9.59 %	122	393	31.04 %	289	651	44.39 %
Fujian				1	1	100.00 %	20	24	83.33 %	55	66	83.33 %
Gansu							1	2	50.00 %	1	5	20.00 %
Guangdong	4	6	66.67 %	11	14	78.57 %	65	109	59.63 %	221	311	71.06 %
Guangxi				2	2	100.00 %	3	20	15.00 %	15	23	65.22 %
Guizhou							6	14	42.86 %	15	37	40.54 %
Hainan								1	0.00 %	2	4	50.00 %
Hebei					1	0.00 %	5	12	41.67 %	17	46	36.96 %
Henan		1	0.00 %		2	0.00 %	8	9	88.89 %	57	78	73.08 %
Heilongjiang				2	4	50.00 %	1	14	7.14 %	3	28	10.71 %
Hubei				3	3	100.00 %	24	38	63.16 %	75	126	59.52 %
Hunan		2	0.00 %	1	17	5.88 %	13	37	35.14 %	56	92	60.87 %
Jilin					3	0.00 %	4	10	40.00 %	9	26	34.62 %
Jiangsu				13	17	76.47 %	107	175	61.14 %	327	511	63.99 %
Jiangxi										6	15	40.00 %
Liaoning				5	5	100.00 %	4	19	21.05 %	28	69	40.58 %
Inner Mongolia							1	1	100.00 %	2	10	20.00 %
Ningxia										5	7	71.43 %
Shandong					1	0.00 %	19	22	86.36 %	136	170	80.00 %
Shanxi				1	1	100.00 %	5	13	38.46 %	7	27	25.93 %
Shaanxi							4	16	25.00 %	34	63	53.97 %
Shanghai	7	13	53.85 %	13	16	81.25 %	52	73	71.23 %	154	220	70.00 %
Sichuan				4	6	66.67 %	18	31	58.06 %	33	73	45.21 %
Tianjin				3	6	50.00 %	3	14	21.43 %	148	196	75.51 %
Xinjiang								1	0.00 %	3	12	25.00 %
Yunnan							18	19	94.74 %	13	28	46.43 %
Zhejiang				6	7	85.71 %	34	52	65.38 %	103	167	61.68 %
Chongqing	3	3	100.00 %	9	10	90.00 %	36	42	85.71 %	81	106	76.42 %

Shanghai, and Chongqing have engaged in IURC in China's NEV industry from 2004 to 2007, where the proportion of intraprovincial collaborations is higher (i.e., over 50 %) in Chongqing, Shanghai, and Guangdong, while the remaining provinces primarily engage in interprovincial collaboration. After 2007, the numbers of patents and collaborations soar drastically due to government policies and the market.

Based on the proportion of intraprovincial collaborations and their changes in three stages between 2008 and 2019, the provinces engaging in IURC in China's NEV industry can be classified into three categories. The first category of provinces consists of those whose proportion of intraprovincial collaborations has always been greater than 50 %, including Fujian, Guangdong, Hubei, Jiangsu, Shanghai, Zhejiang, and Chongqing. This demonstrates that these provinces attach greater importance to intraprovincial collaboration, while regional boundaries critically impact IURC in the NEV industry among these provinces. Notably, the proportion of intraprovincial collaborations in these provinces exhibits a downward trend. The second category of provinces comprises those whose proportion of intraprovincial collaborations has always been less than 50 %, including Beijing, Hebei, and Jilin, which suggests that these provinces attach greater importance to interprovincial collaboration. Simultaneously, the proportion of interprovincial collaborations in these provinces demonstrates a growing trend. The third category of provinces involves those with significant changes in their proportion of intraprovincial collaborations, including Gansu, Heilongjiang, Liaoning, Inner Mongolia, Shanxi, Sichuan, and Yunnan. Specifically, the proportion of intraprovincial collaborations in these provinces drops from over 50 % to less than 50 %, indicating that these provinces have shifted their focus from intraprovincial to interprovincial collaboration.

4.2. Evolution of interprovincial collaboration

4.2.1. Overall characteristics of collaborative networks

As illustrated in Table 7, with rising market demand and China's growing focus on the NEV industry, an increasing number of provinces have taken part in the IURCN in the NEV industry as evidenced by an increase in the number of nodes from seven in the first stage to 31 in the fourth stage. Specifically, all the provinces in China, except Macao, Hong Kong, and Taiwan, have participated in the network. At the same time, the number of relationships increases from eight to 234, while network density increases from 0.096 to 0.252, indicating that the number of provincial industry-university-research collaborations in China's NEV industry is on the rise, accompanied by gradually closer innovation ties. According to the results for average path length and clustering coefficient, the clustering coefficient increases from 0.131 during the period between 2004 and 2007 to 0.559 during the period between 2016 and 2019, while the average path length decreases from 2.043 to 1.845. The previously mentioned trend of higher clustering coefficient but smaller average path length observed in these results suggests that the IURCN in China's NEV industry possesses small-world characteristics, which are also diminishing over time.

4.2.2. Spatial characteristics of collaborative network

To conduct an in-depth analysis of the spatial characteristics of provincial IURC in the NEV industry, this study develops a "breadth-depth" two-dimensional matrix. Specifically, breadth refers to the centrality of the provincial network, that is, the number of provinces the province collaborates with, where the higher the number of provinces it collaborates with, the broader the knowledge source possessed by the province; whereas depth refers to the weighted degree of the provincial network, that is, the number of collaborations the province establishes with other provinces, where the higher the number of such collaborations, the more frequent the province engages in IURC. As the breadth and depth of IURC are measured in terms of average value, the provincial IURCN in China's NEV industry can be classified into four categories: high depth-high breadth (HH), high breadth-low depth (HL), low breadth-high depth (LH), and low breadth-low depth (LL) networks.

Table 7
Overall characteristics of the provincial IURCN in the NEV industry.

Indicator	2004–2007	2008–2011	2012–2015	2016–2019
Number of nodes	7	19	29	31
Number of relationships	8	33	123	234
Network density	0.190	0.096	0.151	0.252
Average degree	1.143	1.737	4.241	7.548
Average path length	2.043	2.572	2.056	1.845
Clustering coefficient	0.131	0.250	0.374	0.559

As seen in Figs. 1 and 2, only two quadrants are occupied in the period from 2004 to 2007. In terms of out-degree, both Beijing and Shanghai appear in the HH quadrant as these two provinces engage in in-depth IURC with more than two provinces; hence, both provinces occupy an integral position in the network. In contrast, Hunan, Guangdong, and Henan are in the LL quadrant. As far as in-degree is concerned, the depth of collaboration in Guangdong and Hunan is the most significant within the HH quadrant, while Fujian, Chongqing, Shanghai, and Beijing are in the LL quadrant.

In terms of out-degree, Beijing has always stood out from the rest as it engages in in-depth IURC with most provinces in the network. While Hunan between 2008 and 2011, Jiangsu, Guangdong, and Hunan between 2012 and 2015, as well as Jiangsu, Guangdong, Zhejiang, Shanghai, Anhui, and Hubei between 2016 and 2019 surpass other provinces in terms of breadth and depth of collaboration, the number of provinces in the HH quadrant exhibits an upward trend from year to year, suggesting that an increasing number of provinces are attaching greater importance and proactively engaging in industry-university-research innovation collaboration in the NEV industry. During the period from 2016 to 2019, Hunan focuses on a shift to the breadth of collaboration.

As far as in-degree is concerned, Guangdong and Beijing have always stood out from the rest as they have been at the core of the network for a long period of time. Since Hunan and Sichuan between 2008 and 2011, Jiangsu between 2012 and 2015, as well as Jiangsu, Shanghai, Hubei, Zhejiang, Shaanxi, Shandong, and Hebei between 2016 and 2019 surpass other provinces in terms of breadth and depth of collaboration, these provinces are in the HH quadrant. Meanwhile, Liaoning, Shanghai, and Hebei between 2008 and 2011, Shanghai between 2012 and 2015, as well as Sichuan, Jilin, Hunan, and Henan between 2016 and 2019 demonstrate a greater breadth of collaboration but lack the depth of collaboration, so these provinces are in the HL quadrant. However, Sichuan and Hunan focus on a shift to the breadth of collaboration during the period from 2016 to 2019. No province is in the LH quadrant, while the remaining provinces are marginal provinces located in the LL quadrant.

In the early stage of IURC in the NEV industry, Beijing, as the capital city that is located at the top of knowledge innovation across the country, drove other provinces to engage in collaboration through knowledge diffusion. With growing market demand and the deepening of research, an increasing number of provinces became deeply involved in the collaborative network; however, the core of the network was still primarily made up of economically developed provinces, while economically backward provinces were located at the edge of the network.

This study visualizes the IURCN in China's NEV industry using ArcGIS, and grades the weights of network ties based on the natural breakpoint method. As illustrated in Fig. 3, the only strongest tie between 2004 and 2007 is Beijing → Guangdong, while the only strong tie in this stage is Beijing → Hunan. Overall, spatial relationships are scarce in the network during this period. From 2008 to 2011, Hunan → Beijing, Beijing → Hunan, Beijing → Jiangsu, Beijing → Sichuan, Hunan → Guangdong, and Jiangsu → Hunan make up the strongest ties in the network, whereas the strong tie is replaced by Beijing → Shaanxi,

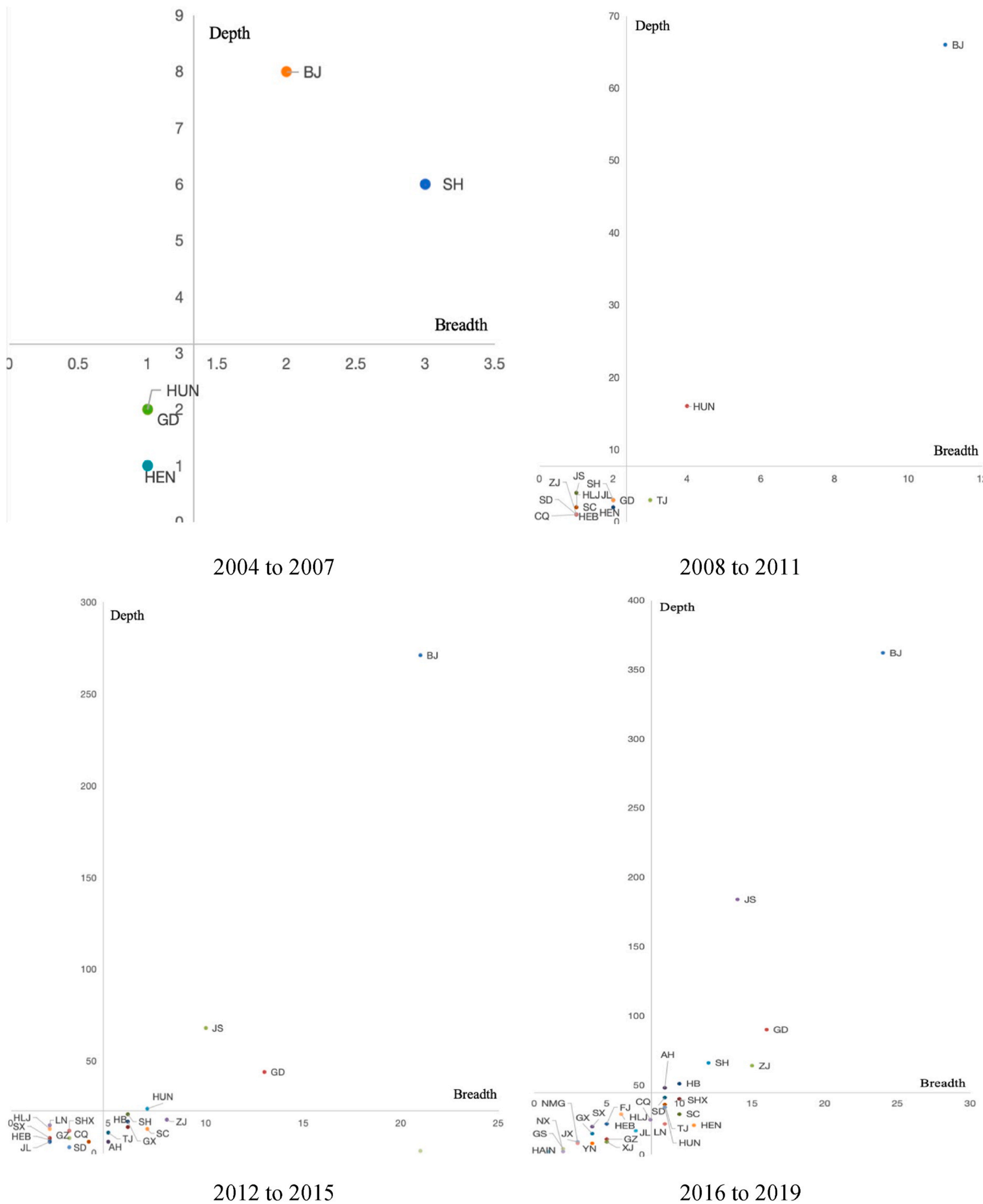

Fig. 1. Out-degree distribution of the IURCN in China's NEV industry.
Note: Beijing = BJ; Guangdong = GD; Jiangsu = JS; Shanghai = SH; Hubei = HB; Shaanxi = SHX; Shandong = SD; Henan = HEN; Zhejiang = ZJ; Sichuan = SC; Hebei = HEB; Hunan = HUN; Jilin = JL; Anhui = AH; Tianjin = TJ; Chongqing = CQ; Heilongjiang = HLJ; Jiangxi = JX; Liaoning = LN; Fujian = FJ; Guangxi = GX; Guizhou = GZ; Gansu = GS; Yunnan = YN; Hainan = HAIN; Inner Mongolia = NMG; Shanxi = SX; Tibet = XZ; Ningxia = NX; Qinghai = QH; Xinjiang = XJ. The same applies hereinafter.

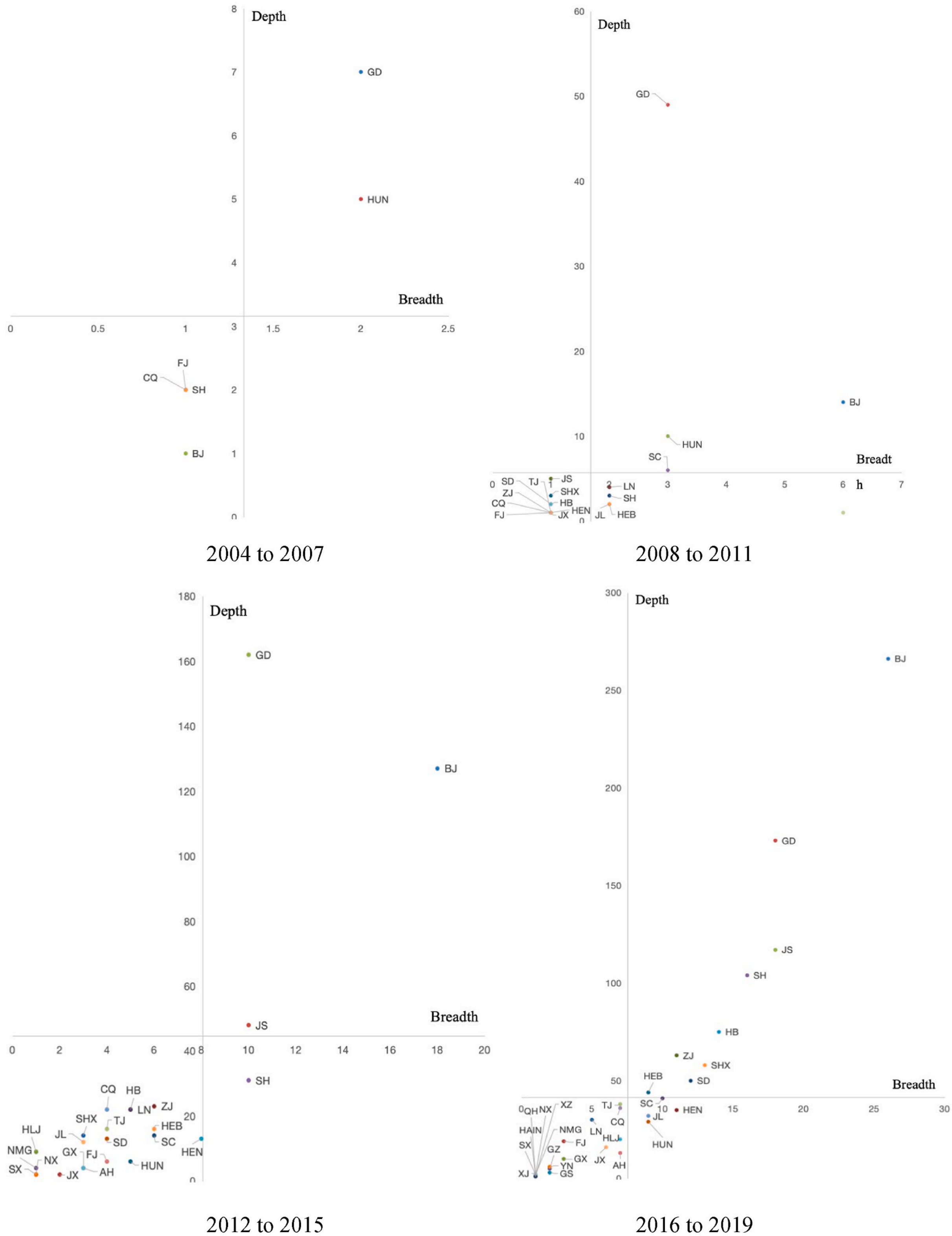

Fig. 2. In-degree distribution of the IURCN in China's NEV industry.

demonstrating that network ties are becoming closer. Building on the network ties in the previous stage, the strongest ties and strong ties continue to expand and spread to the central region during the period between 2012 and 2015. From 2016 to 2019, the strongest ties in the network are basically stable, while strong ties have spatially covered the central and western regions; however, general and weak ties are gradually weakening. Overall, the IURCN in China's NEV industry has developed from a single core into a balanced network, where most provinces in the country have formed a more complete IURCN. Specifically, the network densities in the eastern and central regions are significantly higher than that in the western region, while the integration of networks in the Beijing-Tianjin-Hebei region and the Yangtze

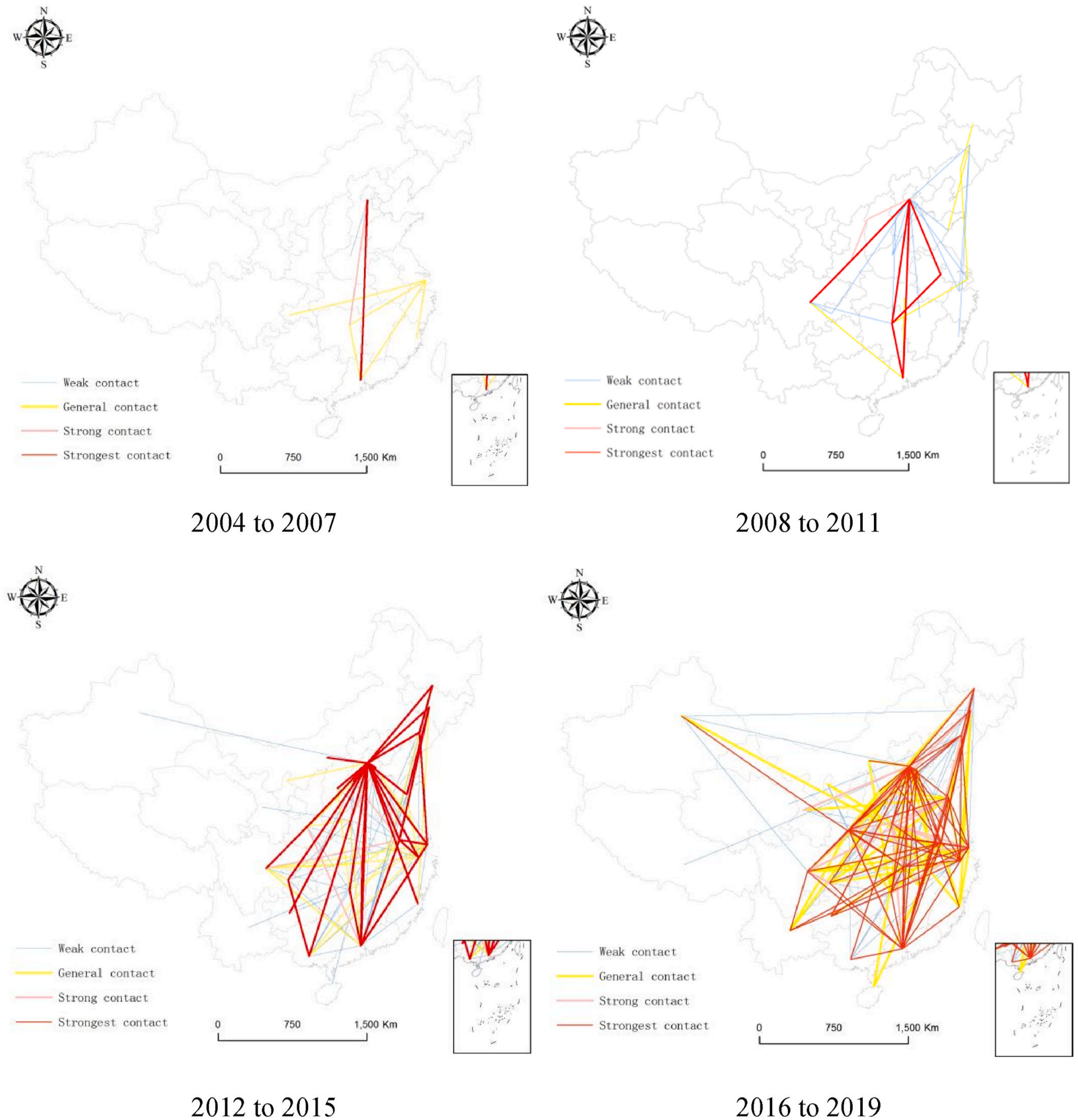

Fig. 3. Spatial relationships in the IURCN in China's NEV industry.

Table 8
Analysis of the factors influencing the provincial IURCN in China's NEV industry.

Indicator	2004–2007		2008–2011		2012–2015		2016–2019	
	Q-value	p-value	Q-value	p-value	Q-value	p-value	Q-value	p-value
Economic base (GDP per capita)	0.449	0.172	0.443	0.193	0.571	0.015	0.541	0.066
Level of openness to the outside world (Total imports and exports of foreign-invested firms)	0.274	0.789	0.355	0.194	0.585	0.028	0.448	0.063
Government dominance (Local government science and technology expenditure)	0.739	0.000	0.393	0.068	0.669	0.021	0.422	0.322
Market dominance (Total retail sales of consumer goods)	0.244	0.201	0.365	0.061	0.557	0.006	0.457	0.129
Regional science and technology finance (Technology market turnover)	0.471	0.620	0.346	0.811	0.544	0.291	0.784	0.000
Entrepreneurial spirit (R&D expenses of industrial firms above designated size)	0.192	0.555	0.279	0.442	0.617	0.028	0.615	0.011

River Delta is gradually intensifying.

5. Factors influencing the provincial IURCN in China's NEV industry

5.1. Selection of influencing factors of collaborative network

Drawing on scholars' views in the past [26,28–31,58–61], this study selects the following factors as explanatory variables while using the centrality of provincial collaborative network as the response variable based on the principles of comprehensiveness and data availability: GDP per capita, which characterizes economic base; total imports and exports of foreign-invested firms, which characterizes the level of openness to the outside world; local government science and technology expenditure, which characterizes government dominance; total retail sales of consumer goods, which characterizes market dominance; technology market turnover, which characterizes regional science and technology finance; and R&D expenditure of industrial firms above designated size, which characterizes entrepreneurial spirit. Meanwhile, this study calculates the coefficients of correlations between the centrality of the provincial collaborative network and its influencing factors, using GeoDetector, and classifies each variable into four levels using breakpoints in ArcGIS before performing analysis using GeoDetector. Due to the possible existence of endogeneity in these explanatory variables, this study uses the end-of-year data of these variables in each stage. The data of all the influencing factors are sourced from the China Statistical Yearbook of previous years.

5.2. Analysis of influencing factors of collaborative network

As illustrated the analysis results obtained using GeoDetector in Table 8, the effect of each influencing factor varies from one stage to another. Government dominance and market dominance play an important role in IURC in China's NEV industry in the early stage, but their effects become insignificant in the later stage. The development of the NEV industry is an important track for China to overtake other countries in the automotive industry. To hasten the development of the NEV industry, the various policies and tax incentives implemented by the central and provincial governments, along with strong demand in the domestic market, have resulted in the significant effects of these two factors. As the NEV industry continues to mature, excessive government intervention and market orientation may lead to profit-seeking behavior in the market and thus exacerbate competition; however, excessive government intervention such as financial subsidies will also result in overcapacity and become detrimental to fair competition in the market. Due to these reasons, these two factors have no significant effect in the later stage.

Under the influence of both government and market dominance in the early stage, the economic base, level of openness to the outside world, regional science and technology finance, and entrepreneurial spirit of provinces in the later stage begin to have a significant effect. Provinces with a better economic base enjoy a higher status in IURC in the NEV industry primarily because the economic base is an important safeguard for the development of the NEV industry. Specifically, the better the economic base, the more beneficial it is to the development of the industry and the improvement of the new energy industry system. Investments from foreign-invested firms in the domestic market can not only bring cutting-edge technology and experience but also improve the entire industry chain and drive the optimal allocation of resources and knowledge between provinces, thereby enhancing the coordination and efficiency of the provincial IURCN in the NEV industry. To promote the development of the domestic NEV industry, China brought in Tesla, the global leader in electric vehicles, in 2012, which in turn prompted local car manufacturers to break away from the policy-dependent path and confront the technology gap, thus accelerating the upgrading of the domestic NEV industry. The higher the level of financial technology in the region, the stronger the entrepreneurial spirit, and thus, the more significant the effect of this factor on promoting the vitality of the IURCN in the regional NEV industry.

6. Conclusions and policy implications

6.1. Conclusions

To illustrate the structural characteristics of the IURCN in the NEV industry, this study analyzes the IURCI network in China's NEV industry using various software programs such as Gephi and ArcGIS based on the data provided in global patent database incoPat. The following conclusions can be drawn from this study.

(1) The highest number of patents obtained through IURC is observed in NEV device and accessory manufacturing, whereas the number of new entrants is significantly higher than the number of incumbents albeit in a downward trend, suggesting that the creation of new collaborative relationships among new entrants has always been the main phenomenon in the process of building an industry-university-research innovation system in the NEV industry. Since private firms have long been playing an integral role in IURC in the NEV industry, the number of private firms in the network exhibits an upward trend on a yearly basis, while the dual heterogeneous connection mode is the primary connection mode in IURC.

(2) On the subject of intraprovincial collaboration in the IURCN in China's NEV industry, Fujian, Guangdong, Hubei, and Jiangsu attach greater importance to intraprovincial collaboration, while the opposite circumstance is observed in Beijing, Hebei, and Jilin. However, Gansu, Heilongjiang, and Liaoning are gradually shifting their focus from intraprovincial to interprovincial. As for interprovincial collaboration, network ties are gradually becoming closer while the small-world characteristics of the network are gradually weakening. Meanwhile, Beijing is at the top when it comes to the breadth and depth of collaboration, be it in terms of out-degree or in-degree. Furthermore, the core of the network is dominated by economically developed provinces, whereas economically backward provinces are located at the edge of the network.

(3) According to the results obtained using GeoDetector, the effects of the economic base, level of openness to the outside world, government dominance, market dominance, regional science and technology finance, and entrepreneurial spirit vary from one stage to another. Government dominance and market dominance in the early stage play an integral role in IURC in China's NEV industry. Under the influence of both government and market dominance in the early stage, the economic base, level of openness to the outside world, regional science and technology finance, and entrepreneurial spirit of provinces in the later stage begin to have a significant effect.

6.2. Policy implications

(1) From the viewpoint of IURC in the NEV industry by region, provinces located in the HH quadrant should make continuous efforts to deepen industry-university-research collaborative innovation in the NEV industry, boost the influence of such collaboration, and stabilize their position in the network. For provinces in the LL quadrant, their breadth and depth of collaboration are lower than the average values, while the after-effects of developing IURCI in these provinces are less sufficient. Therefore, local governments in these provinces have to vigorously promote such collaboration through a series of measures, including building an IURC platform for the NEV industry, establishing a reward and punishment system, lowering

collaboration costs, improving collaborative innovation policies, optimizing the IURCI environment, and opening up the channels for cross-regional collaborative innovation.

(2) Regarding the industry-university-research innovation stage, the following results from the regression demonstrate the influencing factors. First, present studies on IURCI in the NEV industry have demonstrated that excessive government intervention and free market dominance are no longer conducive to the development of such innovation. Hence, there is a need for proper adjustment and balance between the collaborating entities to ensure that IURCI in the NEV industry is efficient and sustainable. Second, it is necessary to inculcate the spirit of innovation in entrepreneurs and thus increase the level of regional science and technology finance. The spirit of innovation among entrepreneurs and the level of regional science and technology finance are two factors that can significantly influence IURCI in the NEV industry. On the one hand, local governments can subtly inculcate the spirit of innovation in entrepreneurs through the implementation of relevant training programs and promotional campaigns; on the other hand, local governments need to formulate corresponding strategies for developing regional science and technology finance in combination with their own characteristics to accelerate the integration and coordinated development of science and technology finance.

(3) From the viewpoint of nodal and organizational characteristics and connection modes in the IURCN, there are less collaborations between incumbents and new entrants. While the existing path dependence has caused certain barriers to collaboration for new entrants, there is a need to improve competitiveness in various areas such as infrastructure and talent development, bolster the technological innovation capabilities of firms, and proactively overcome barriers to collaboration using innovation platforms. Since the dual heterogeneous connection mode is the primary connection mode in IURC, local governments need to establish an IURC mechanism aimed at keeping up with demand in the NEV market, improving collaboration policies for heterogeneous patent applicants, and mitigating problems related to poor connection, thereby facilitating the efficient operation of the industry-university-research collaborative network in the NEV industry.

6.3. Research limitations and outlook

(1) Despite the intricate and complex industry-university-research collaborations between different countries against the backdrop of globalization, this study only includes China in its scope of research but lacks investigations into innovation collaboration on charging piles for new energy vehicles between different countries from a global-local perspective. In addition, IURC in prefecture-level cities and counties is still pending research.

(2) As regards research on IURC in the NEV industry, this study conducts analyses solely from a patent collaboration perspective, which can only demonstrate one side of innovation collaboration. The pattern of evolution of the IURCN in the NEV industry can be comprehensively investigated in the form of journal paper collaboration and questionnaire survey in future studies.

(3) On the factors influencing the IURCN in the NEV industry, this study only analyzes the general characteristics of the patterns of this network. Future studies can dissect and analyze in depth the factors influencing the network in various areas such as policy environment and industry development characteristics from the viewpoints of multi-dimensional proximity and the development of each network node.

All authors have read and agreed to the published version of the manuscript.

CRediT authorship contribution statement

Xiaoping Wang: Data curation, Conceptualization, Writing – original draft, Writing – review & editing. **Liping Qiu:** Data curation, Investigation, Writing – review & editing. **Feng Hu:** Conceptualization, Resources, Supervision. **Hao Hu:** Formal analysis, Software, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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